CPEN 416

Lecture 13: QFT and quantum phase estimation

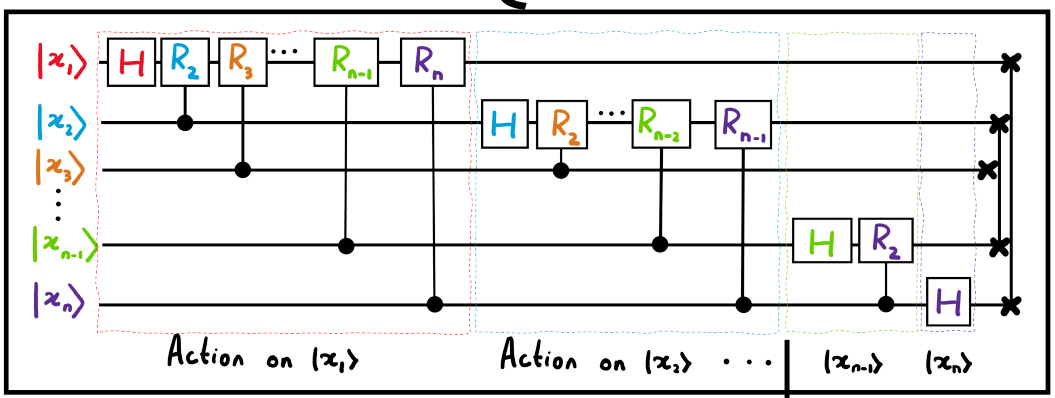
Tuesday 27 October 2026

Announcements

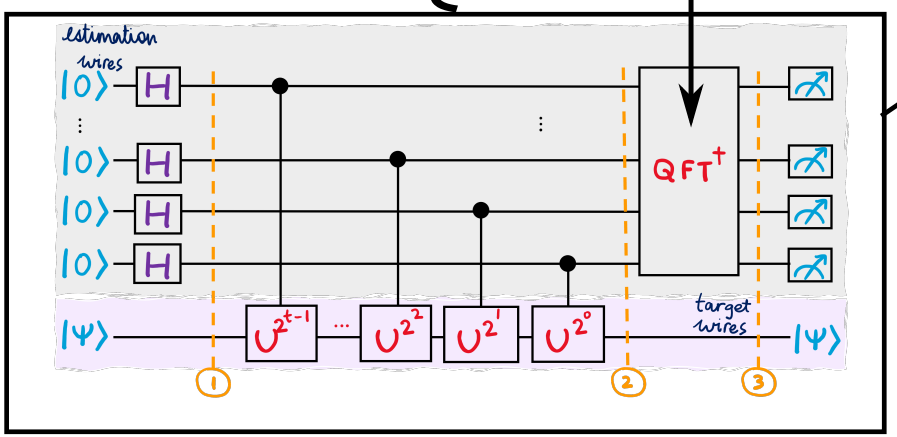
- Quiz 6 today
- Tutorial Assignment 2 due Friday 23:59
- Project midterm checkpoint due Monday; sign up for appointment time in Canvas tomorrow

Reminder: where are we going?

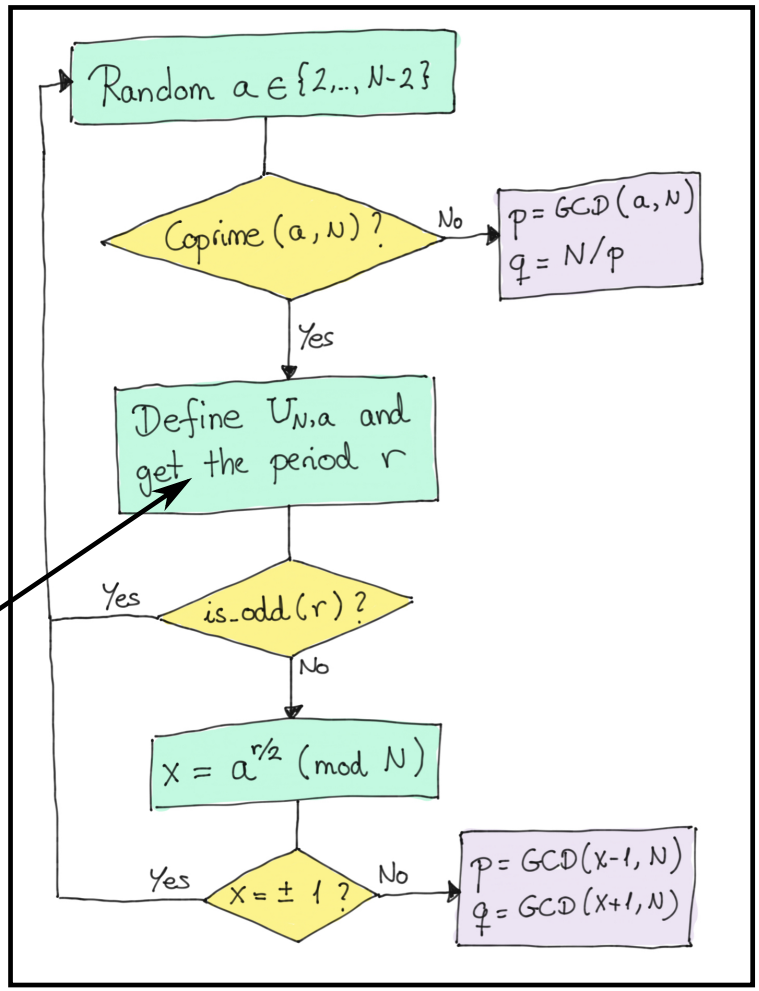
1. QFT



2. QPE



3. Shor



Last time

We introduced the quantum Fourier transform, which is analogous to the inverse discrete Fourier transform.

$$QFT|x\rangle = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} \omega^{xk} |k\rangle$$

$$QFT = \frac{1}{\sqrt{N}} \begin{pmatrix} 1 & 1 & 1 & \cdots & 1 \\ 1 & \omega & \omega^2 & \cdots & \omega^{N-1} \\ 1 & \omega^2 & \omega^4 & \cdots & \omega^{2(N-1)} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & \omega^{N-1} & \omega^{2(N-1)} & \cdots & \omega^{(N-1)(N-1)} \end{pmatrix}$$

where for n qubits, $N = 2^n$, and $\omega = e^{2\pi i/N}$

Learning outcomes

- Describe the behaviour of the quantum Fourier transform
- Construct a circuit for the quantum Fourier transform and analyze its resource usage
- Outline the algorithmic steps of the QPE subroutine

Quantum Fourier transform: 1 qubit

Quantum Fourier transform: 2 qubits

$$QFT = \frac{1}{2}\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & \omega & \omega^2 & \omega^3 \\ 1 & \omega^2 & \omega^4 & \omega^6 \\ 1 & \omega^3 & \omega^6 & \omega^9 \end{pmatrix}$$

Re-express higher powers of $\omega = e^{\frac{2\pi i}{4}}$:

Quantum Fourier transform: 2 qubits

$$QFT = \frac{1}{2}\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & \omega & \omega^2 & \omega^3 \\ 1 & \omega^2 & 1 & \omega^2 \\ 1 & \omega^3 & \omega^2 & \omega^1 \end{pmatrix}$$

How does this affect the computational basis states?

Quantum Fourier transform: 2 qubits

Quantum Fourier transform: 2 qubits

Quantum Fourier transform: 3 qubits

$$QFT = \frac{1}{\sqrt{8}} \begin{pmatrix} 1 & 1 & 1 & 1 & 1 & \cdots \\ 1 & \omega & \omega^2 & \omega^3 & \omega^4 & \cdots \\ 1 & \omega^2 & \omega^4 & \omega^6 & 1 & \cdots \\ 1 & \omega^3 & \omega^6 & \omega & \omega^4 & \cdots \\ 1 & \omega^4 & 1 & \omega^4 & 1 & \cdots \end{pmatrix}$$

$$QFT|x_1x_2x_3\rangle = \frac{1}{\sqrt{2}}\left(|0\rangle + e^{2\pi i 0.x_3}|1\rangle\right) \otimes \frac{1}{\sqrt{2}}(|0\rangle + e^{2\pi i 0.x_2x_3}|1\rangle) \otimes \frac{1}{\sqrt{2}}(|0\rangle + e^{2\pi i 0.x_1x_2x_3}|1\rangle)$$

Quantum Fourier transform: n qubits

General expression:

$$
\begin{aligned}
|x\rangle \quad &\rightarrow \quad \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} \omega^{xk} |k\rangle \\
&= \quad \frac{\left(|0\rangle + e^{2\pi i 0.x_n}|1\rangle\right)\left(|0\rangle + e^{2\pi i 0.x_{n-1}x_n}|1\rangle\right) \cdots \left(|0\rangle + e^{2\pi i 0.x_1 \cdots x_n}|1\rangle\right)}{\sqrt{N}}
\end{aligned}
$$

Full mathematical derivation at end of slides.

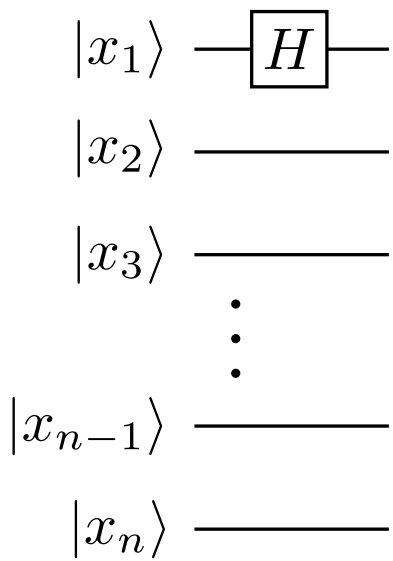
A circuit for the QFT

$$|x\rangle \to \frac{\left(|0\rangle + e^{2\pi i 0.x_n}|1\rangle\right)\left(|0\rangle + e^{2\pi i 0.x_{n-1}x_n}|1\rangle\right)\cdots\left(|0\rangle + e^{2\pi i 0.x_1\cdots x_n}|1\rangle\right)}{\sqrt{N}}$$

Starting with the state

$$|x\rangle = |x_1 \cdots x_n\rangle,$$

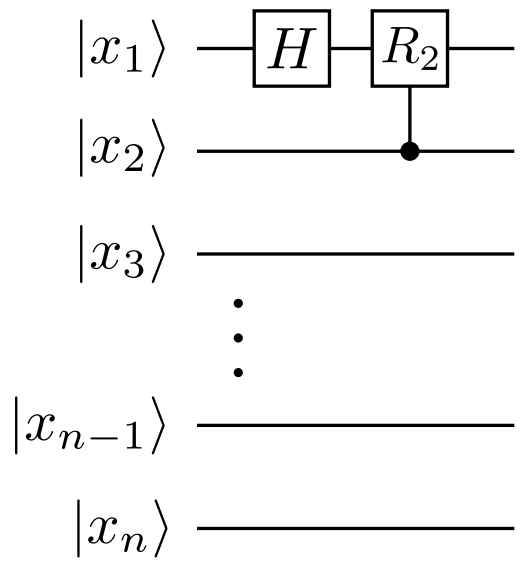
apply a Hadamard to qubit 1:



A circuit for the QFT

$$|x\rangle \rightarrow \frac{\left(|0\rangle + e^{2\pi i 0.x_n}|1\rangle\right)\left(|0\rangle + e^{2\pi i 0.x_{n-1}x_n}|1\rangle\right)\cdots\left(|0\rangle + e^{2\pi i 0.x_1\cdots x_n}|1\rangle\right)}{\sqrt{N}}$$

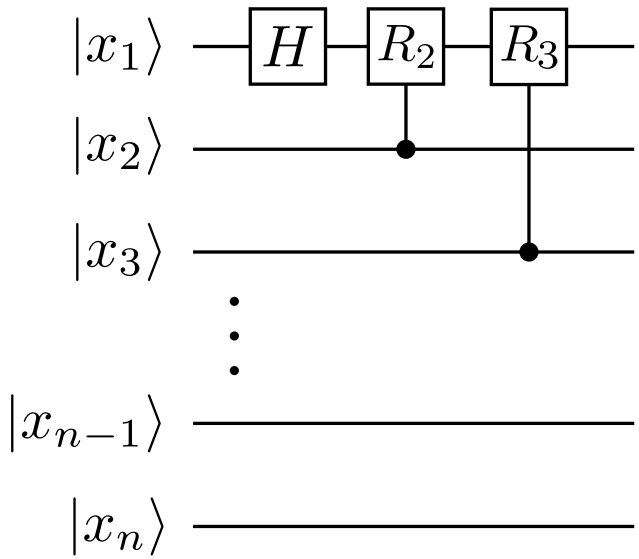
Apply controlled R_2 from qubit $2 \rightarrow 1$



A circuit for the QFT

$$|x\rangle \rightarrow \frac{\left(|0\rangle + e^{2\pi i 0.x_n}|1\rangle\right)\left(|0\rangle + e^{2\pi i 0.x_{n-1}x_n}|1\rangle\right)\cdots\left(|0\rangle + e^{2\pi i 0.x_1\cdots x_n}|1\rangle\right)}{\sqrt{N}}$$

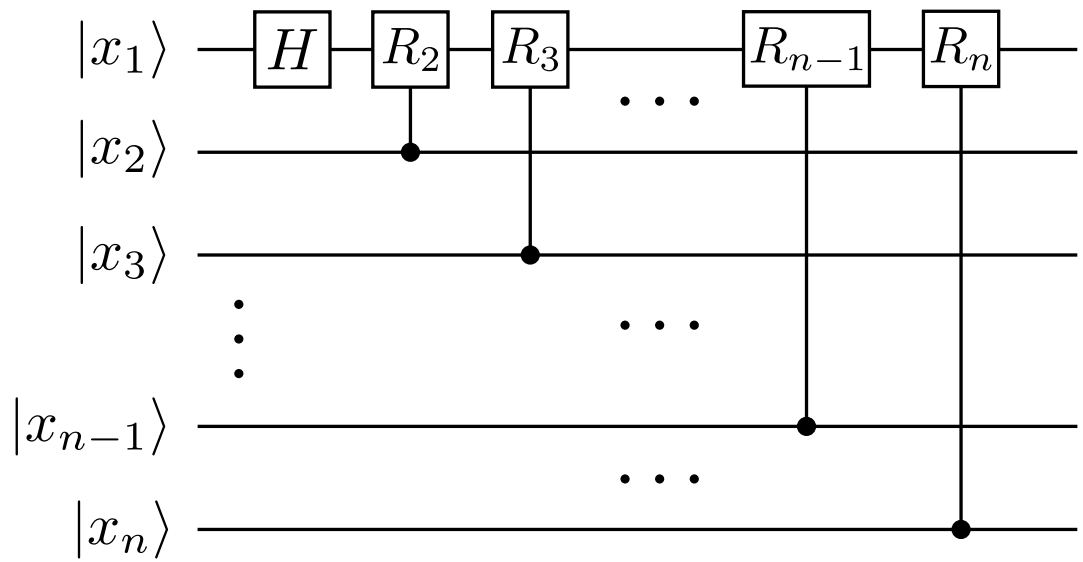
Apply controlled R_3 from qubit $3 \rightarrow 1$



A circuit for the QFT

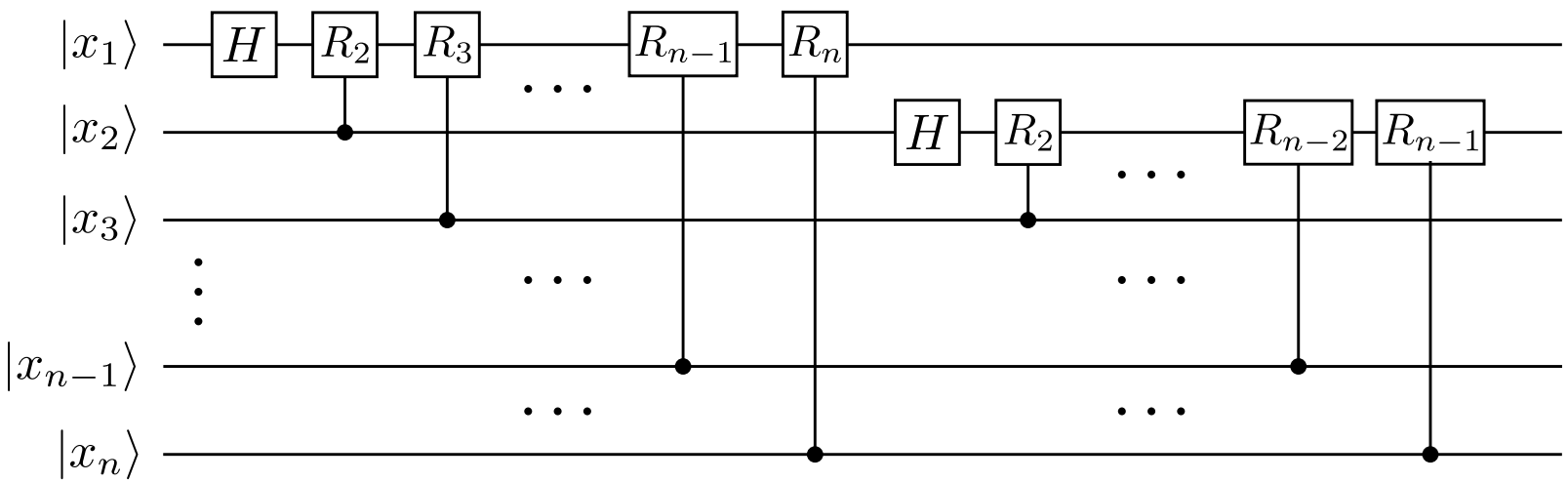
$$|x\rangle \rightarrow \frac{\left(|0\rangle + e^{2\pi i 0.x_n}|1\rangle\right)\left(|0\rangle + e^{2\pi i 0.x_{n-1}x_n}|1\rangle\right)\cdots\left(|0\rangle + e^{2\pi i 0.x_1\cdots x_n}|1\rangle\right)}{\sqrt{N}}$$

Apply a controlled R_4 from $4 \rightarrow 1$, etc. up to the n-th qubit to get



A circuit for the QFT

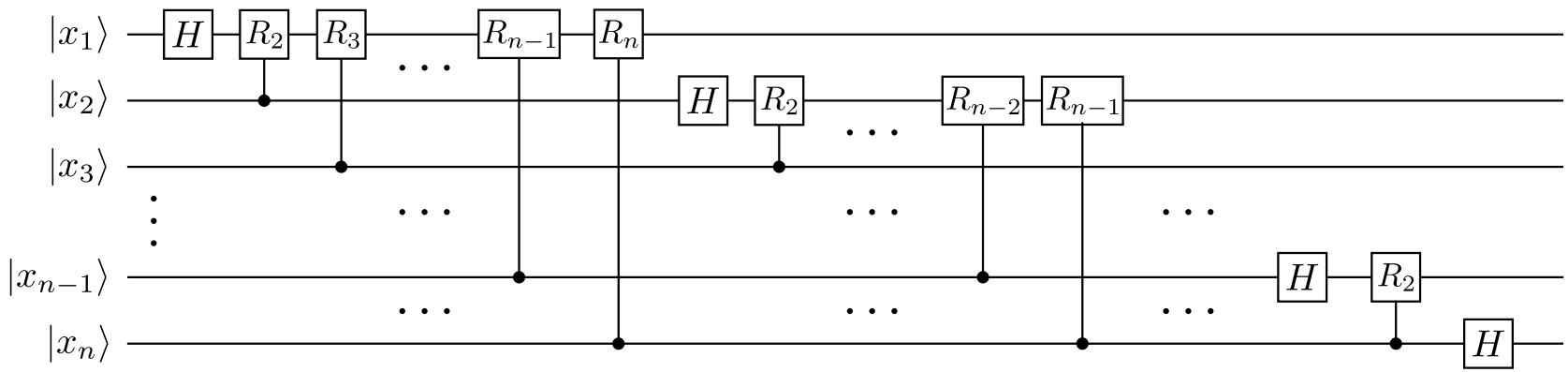
Similar thing with second qubit: apply H then controlled rotations from qubits 3 to n to get



A circuit for the QFT

Do this for all qubits to get that big ugly state from earlier:

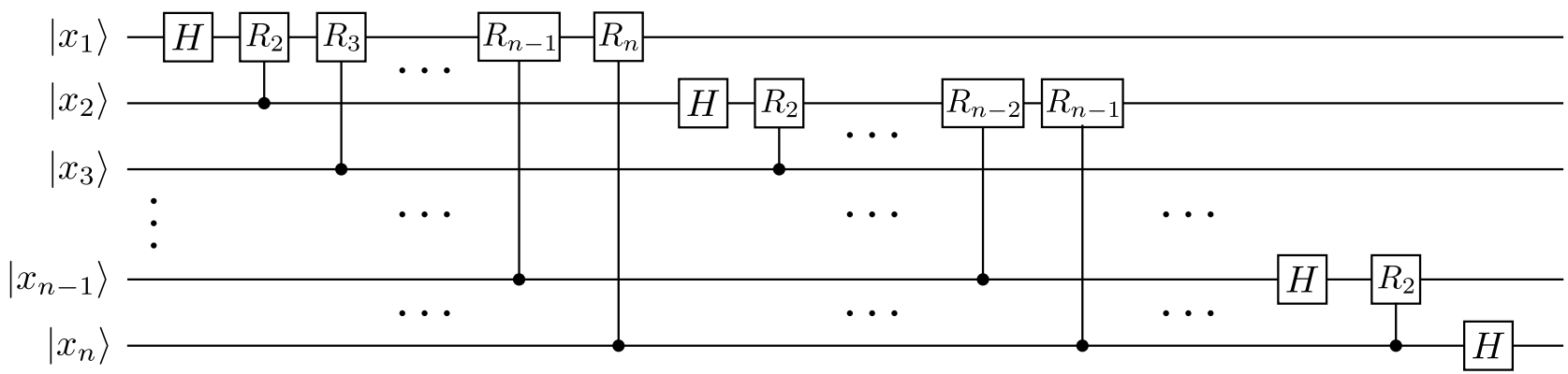
$$|x\rangle \rightarrow \frac{\left(|0\rangle + e^{2\pi i 0.x_n}|1\rangle\right)\left(|0\rangle + e^{2\pi i 0.x_{n-1}x_n}|1\rangle\right)\cdots\left(|0\rangle + e^{2\pi i 0.x_1\cdots x_n}|1\rangle\right)}{\sqrt{N}}$$



Qubit order is backwards - easily fixed with SWAP gates.

Quantum Fourier transform

Exercise: What are the gate counts and depth of this circuit?



Eigenvalues of unitary matrices

Eigenvalues of unitary matrices are complex numbers with magnitude 1, i.e.,

where θ_k is a phase, $|\theta_k| \leq 1$. (See proof at end of slides)

What if we want to *estimate* an unknown θ_k?

Quantum phase estimation

Given unitary U and eigenvector $|k\rangle$, estimate θ_k such that

$$U|k\rangle = e^{2\pi i\theta_k}|k\rangle$$

Must determine:

- How to design a circuit that extracts θ_k
- What precision can we estimate it to
- What to do if we don't know $|k\rangle$ in advance

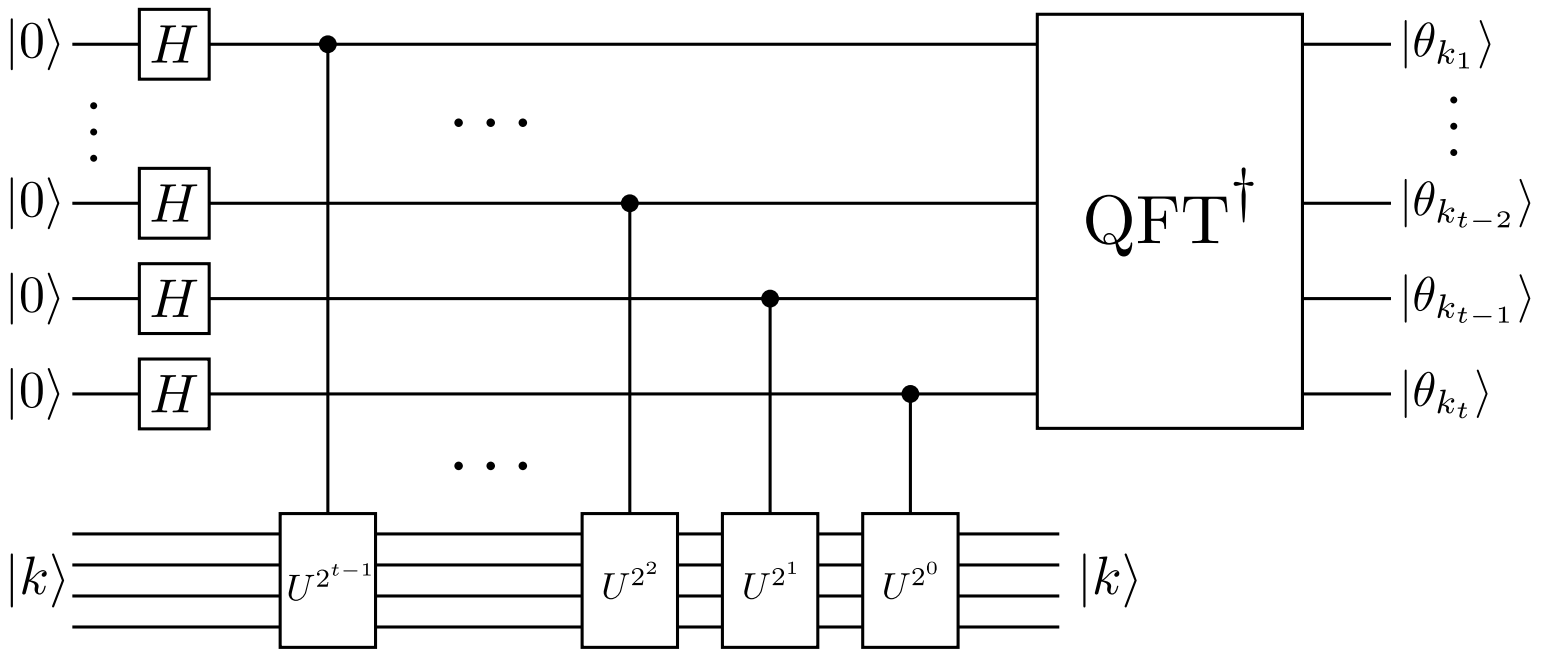
(*You can explore the last two in practice assignment 3!*)

Quantum phase estimation

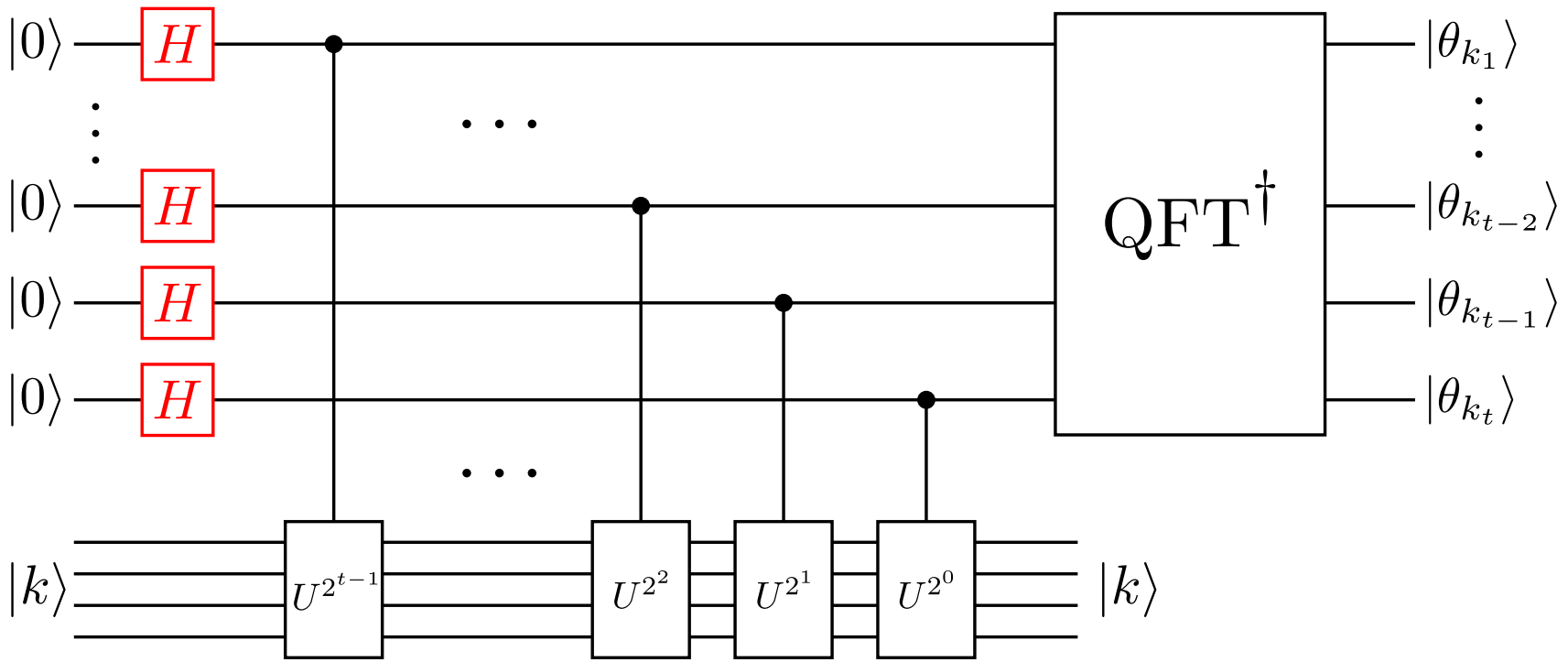
Let U be an n-qubit unitary; $|k\rangle$ is an n-qubit eigenstate.

Assume θ_k can be represented *exactly* using t bits:

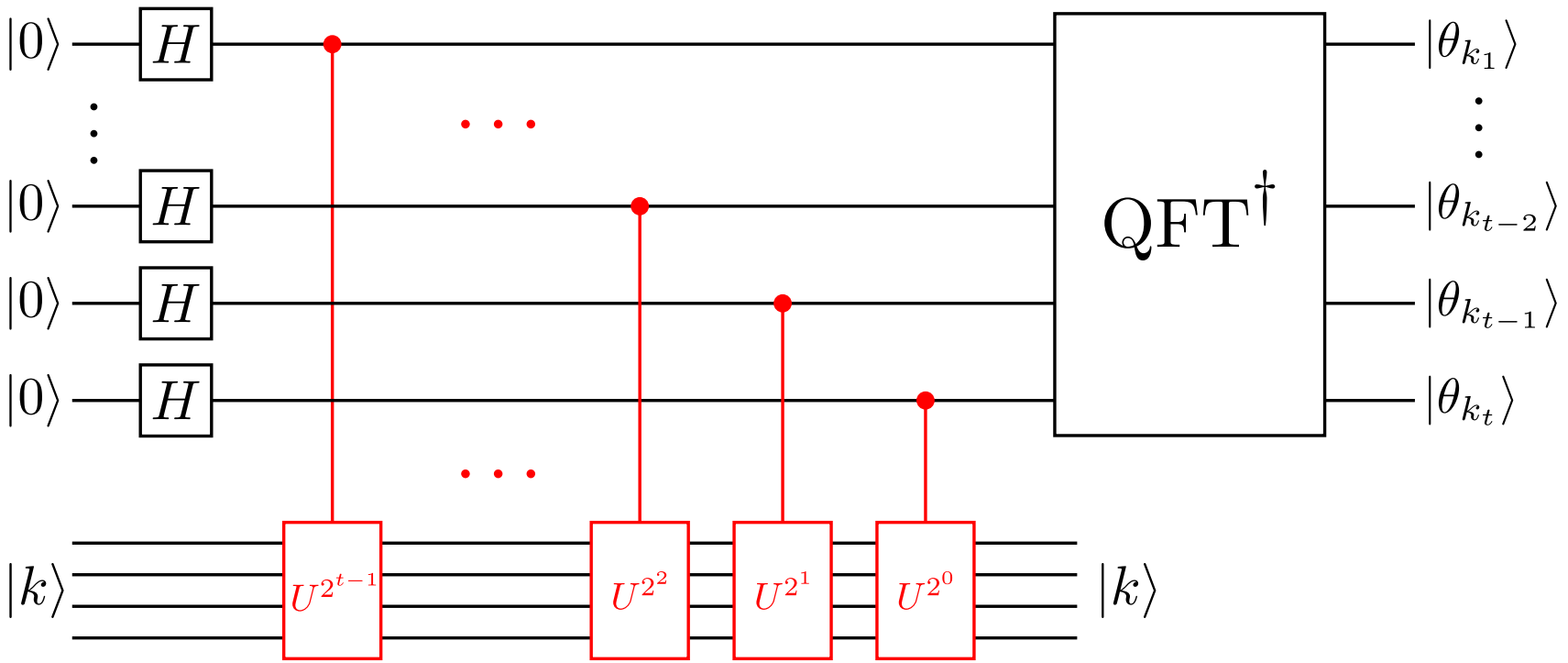
$$\theta_k = 0.\theta_{k_1} \cdots \theta_{k_t}$$



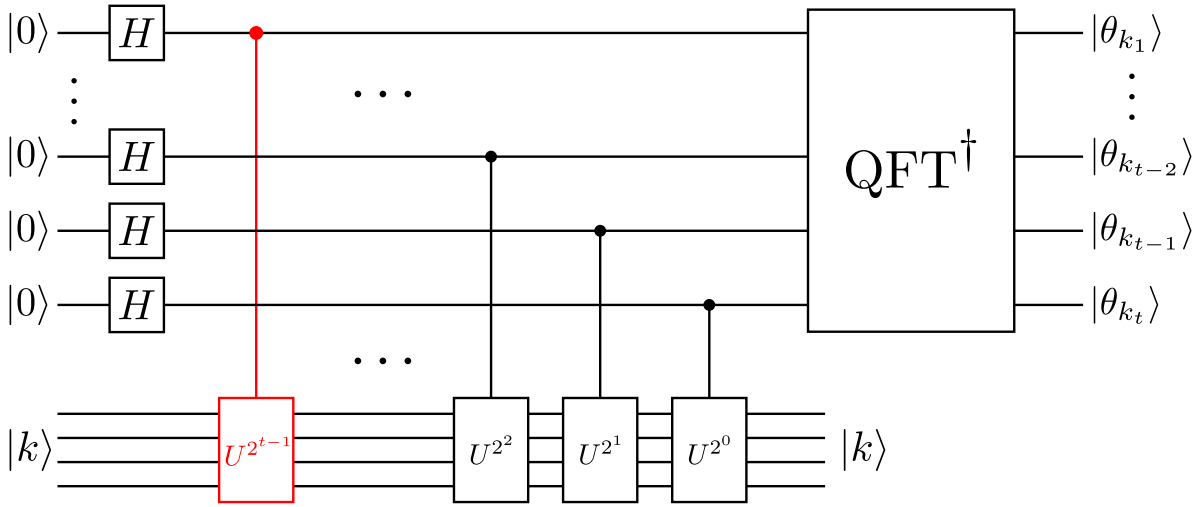
Quantum phase estimation: step 1



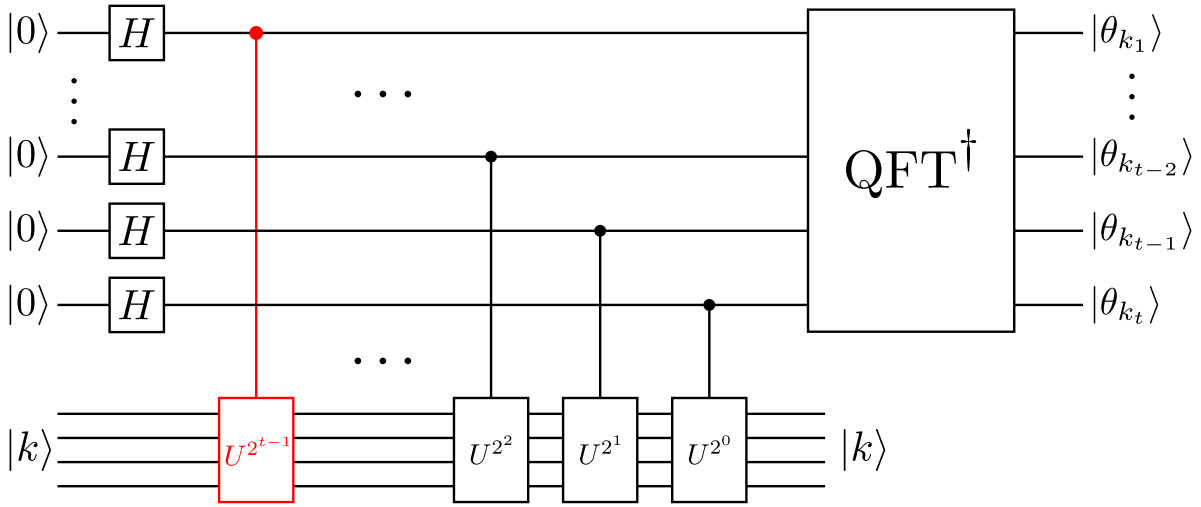
Quantum phase estimation: step 2



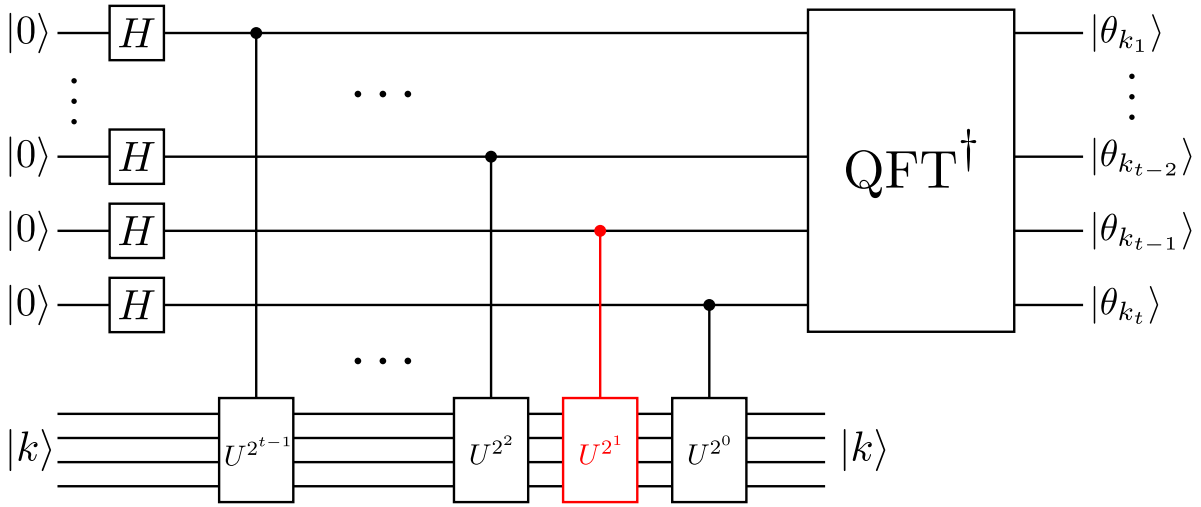
Quantum phase estimation: step 2



Quantum phase estimation: step 2

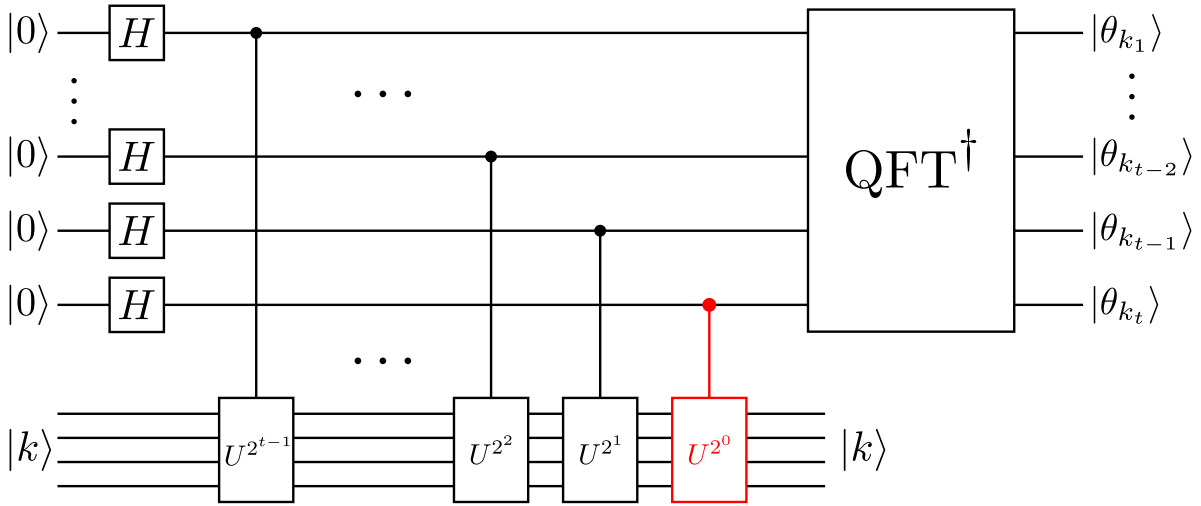


Quantum phase estimation: step 2



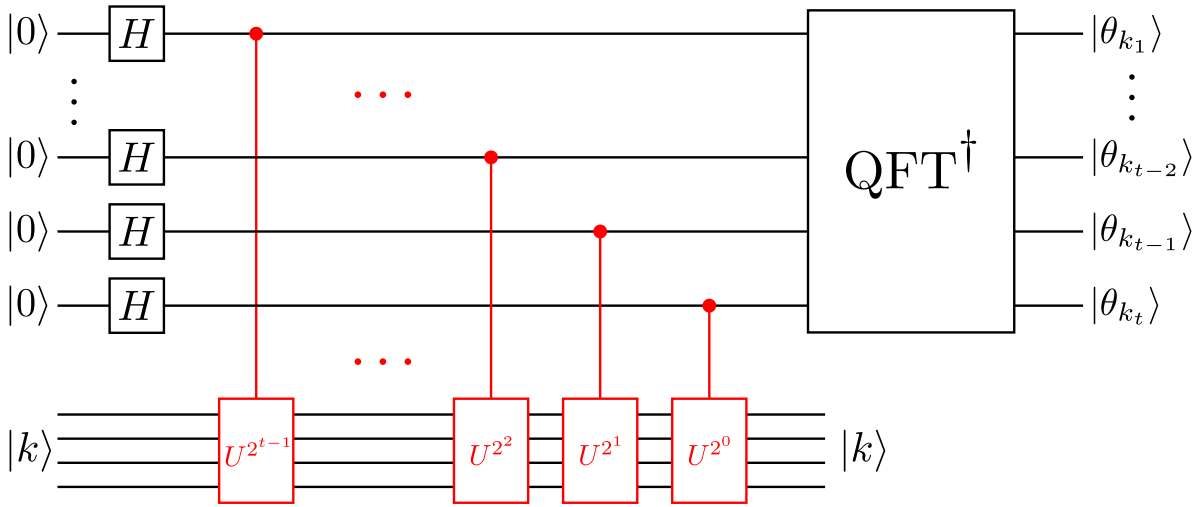
Check second-last qubit (ignore the others)

Quantum phase estimation: step 2



Can show in the same way for the last qubit (ignore others)

Quantum phase estimation: step 2



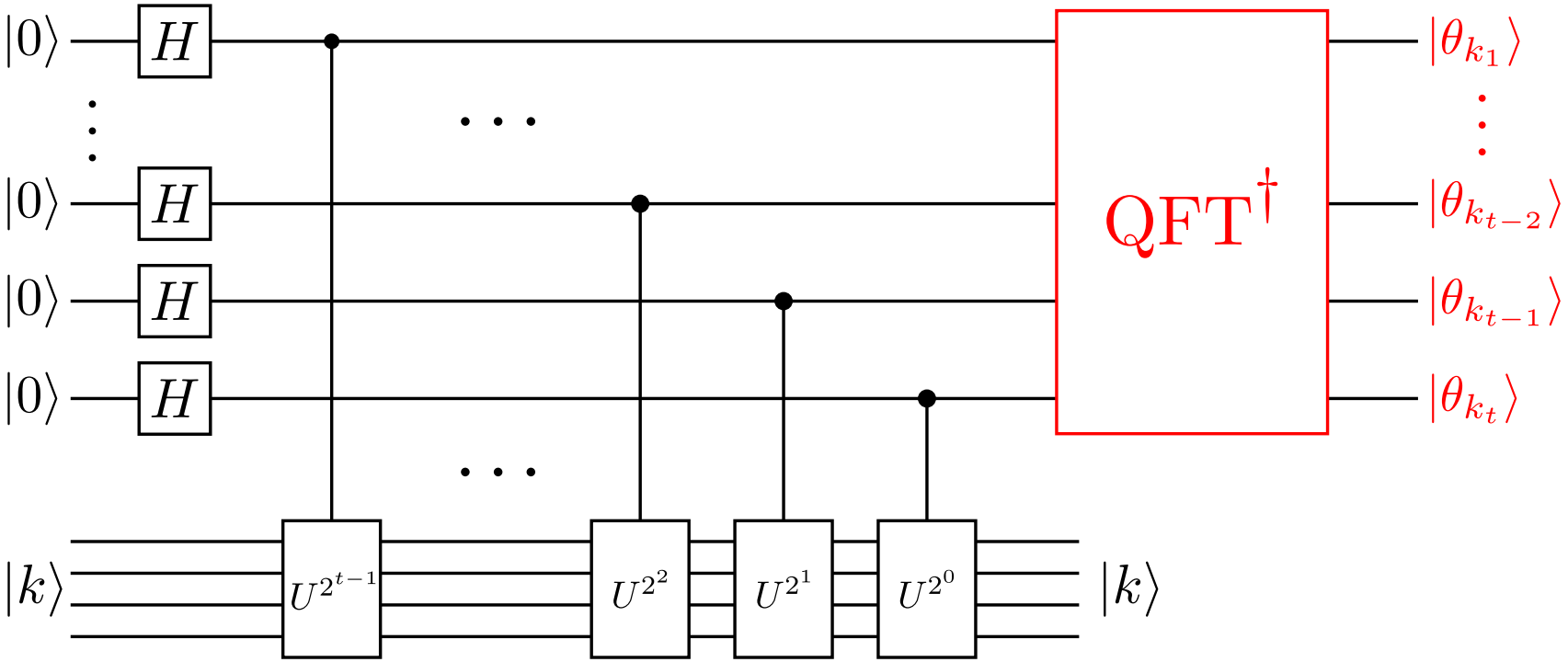
After step 2, we have the state

$$\frac{1}{\sqrt{2}}(|0\rangle + e^{2\pi i 0.\theta_{k_t}}|1\rangle) \cdots \frac{1}{\sqrt{2}}(|0\rangle + e^{2\pi i 0.\theta_{k_2}\cdots\theta_{k_t}}|1\rangle)\frac{1}{\sqrt{2}}(|0\rangle + e^{2\pi i 0.\theta_{k_1}\cdots\theta_{k_t}}|1\rangle)|k\rangle$$

Should look familiar!

Quantum phase estimation: step 3

Measure to learn the bits of θ_k.



Example: QPE for the T gate

Let's apply QPE to estimate the phase of an eigenstate of T: $|1\rangle$.

1. What answer do we expect?
2. How many estimation bits?
3. What does the circuit look like?

Quantum phase estimation

Recall our initial questions... any insights about the second two?

- How to design a circuit that extracts θ_k
- What precision can we estimate it to
- What to do if we don't know $|k\rangle$ in advance

Next time

Content:

- Toward's Shor's algorithm: order finding with QPE

Action items:

1. Finish tutorial assignment 2
2. Work on project (and fill peer assessment)

Recommended reading:

- From today: Codebook QFT, QPE, Nielsen & Chuang 5.1-2
- For next class: Codebook SH, Nielsen & Chuang 5.3.1

Derivation of output state of QFT

We will reexpress k/N in fractional binary notation, then reshuffle and *factor* the output state.

$$
\begin{aligned}
|x\rangle &\rightarrow \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} \omega^{xk} |k\rangle \\
&= \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} e^{2\pi i x (k/N)} |k\rangle \\
&= \frac{1}{\sqrt{N}} \sum_{k_1=0}^{1} \cdots \sum_{k_n=0}^{1} e^{2\pi i x \left(\sum_{\ell=1}^{n} k_\ell 2^{-\ell}\right)} |k_1 \cdots k_n\rangle \\
&= \frac{1}{\sqrt{N}} \sum_{k_1=0}^{1} \cdots \sum_{k_n=0}^{1} \bigotimes_{\ell=1}^{n} e^{2\pi i x k_\ell 2^{-\ell}} |k_\ell\rangle
\end{aligned}
$$

Derivation of output state of QFT

$$
\begin{aligned}
&= \frac{1}{\sqrt{N}} \sum_{k_1=0}^{1} \cdots \sum_{k_n=0}^{1} \bigotimes_{\ell=1}^{n} e^{2\pi i x k_\ell 2^{-\ell}} |k_\ell\rangle \\
&= \frac{1}{\sqrt{N}} \bigotimes_{\ell=1}^{n} \left(\sum_{k_\ell=0}^{1} e^{2\pi i x k_\ell 2^{-\ell}} |k_\ell\rangle \right) \\
&= \frac{1}{\sqrt{N}} \bigotimes_{\ell=1}^{n} \left(|0\rangle + e^{2\pi i x 2^{-\ell}} |1\rangle \right) \\
&= \frac{\left(|0\rangle + e^{2\pi i 0.x_n}|1\rangle\right)\left(|0\rangle + e^{2\pi i 0.x_{n-1}x_n}|1\rangle\right) \cdots \left(|0\rangle + e^{2\pi i 0.x_1 \cdots x_n}|1\rangle\right)}{\sqrt{N}}
\end{aligned}
$$

Eigenvalues of unitary matrices

Fun fact: eigenvalues of unitaries are complex numbers with magnitude 1.

Proof: Let λ_k be the eigenvalue associated with eigenvector $|k\rangle$ of a unitary U:

$$U|k\rangle = \lambda_k|k\rangle$$

Take the conjugate transpose:

$$\langle k| U^\dagger = \langle k| \lambda_k^*$$

Multiply the two sides together:

$$\begin{aligned}\langle k| U^\dagger U|k\rangle &= \langle k| \lambda_k^*\lambda_k|k\rangle \\ \langle k|k\rangle &= |\lambda_k|^2\langle k|k\rangle \\ 1 &= |\lambda_k|^2\end{aligned}$$